

JOAQUÍN CARDONA RUIZ

MSc. Artificial Intelligence | NLP & Deep Learning

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The Netherlands, EU Citizen

RESEARCH INTERESTS

Compositional approaches to language and meaning, and the interplay between symbolic and neural representations in models of language understanding. Interested in how linguistic structure can inform the design and evaluation of AI systems, and in what computational models can reveal about human language and cognition.

EDUCATION

MSc. Artificial Intelligence

Vrije Universiteit Amsterdam & Universiteit van Amsterdam

Sep 2023 – Sep 2025

Amsterdam, The Netherlands

- **Cum Laude;** Weighted average grade of 88%
- **Thesis:** *Examining Algebraic Recombination for Compositional Generalisation:* designed and implemented a non-projective semantic parser to study compositional generalisation in neural models of language. Graded 90%. (Supervisors: Lucia Donatelli, Antske Fokkens).
- **Relevant coursework:** Advanced Machine Learning, Advanced NLP, Applied Text Mining, Comparative Linguistics, Deep Learning, Experimental Design & Data Analysis for AI, Linguistic Research, Machine Learning for NLP, Natural Language Processing.

Software Engineering Long-cycle Degree (BSc. + MSc.)

National University of Technology

Mar 2016 – Apr 2021

Rosario, Argentina

- Focus on Artificial Intelligence, Data Mining, and Statistics

PUBLICATIONS

Examining Algebraic Recombination for Compositional Generalisation

May 2026

🔗 | **Joaquín Cardona Ruiz**, Antske Fokkens, Lucia Donatelli.

Introduced DEAR (Dependency Algebraic Recombination), a compositional semantic parser extending the LEAR model by replacing its constituency-based intermediate representation with an unconstrained dependency structure, enabling the modelling of non-projective syntactic dependencies. Evaluated on SLOG, a structural generalisation benchmark. DEAR achieves non-zero accuracy on all categories — including non-projective constructions where projective baselines score 0% — and substantially outperforms LEAR on center-embedding constructions (0% → 95.9%). Overall accuracy is comparable to the projective baseline (53.1% vs. 54.2%), reflecting a fundamental trade-off between expressive capacity and search-space tractability. Accepted to be published at CMCL 2026 (Workshop on Cognitive Modeling and Computational Linguistics).

PRESENTATIONS

Joaquín Cardona Ruiz, Antske Fokkens, Lucia Donatelli.

Upcoming

Examining Algebraic Recombination for Compositional Generalisation. Poster to be presented at The 15th edition of the Workshop on Cognitive Modeling and Computational Linguistics (CMCL 2026), Palau de Congressos de Palma, Palma de Mallorca (Spain), 16 May 2026.

Irene Girdin, Guillermo Leale, **Joaquín Cardona Ruiz**, Magdalena Robaglio.

Oct 2019

Meaning Disputes in the First Presidential Debates in Argentinian History. Talk at The 15th edition of Rosario's Congress of Socio-cultural Anthropology, Rosario, Argentina, 25 October 2019.

Joaquín Cardona Ruiz, Agustín Delmonti, Joaquín Sánchez.

Nov 2018

Categorization of Text-based Forum Posts Using Neural Networks. Talk at The 6th edition of the National Congress of Software & Information Systems Engineering, Mar del Plata, Argentina, 30 November 2018.

ACADEMIC EXPERIENCE

- Student Researcher**
Vrije Universiteit Amsterdam
Conducted research to expand scope of master’s thesis work for publication at CMCL 2026 by performing additional experiments and broadening comparisons to recent literature.

Oct 2025 – Feb 2026
Amsterdam, The Netherlands
- Teaching Assistant**
Vrije Universiteit Amsterdam
Assisted professor Lucia Donatelli on teaching text corpora to master’s students for Language as Data class at Vrije Universiteit Amsterdam. Provided students with interactive workshop content to learn how to access, manipulate, and draw analytical conclusions from publicly-accessible text corpora.

Nov 2025
Amsterdam, The Netherlands
- Student Researcher**
National University of Technology
Performed research as part of an NLP-focused research group within Software Engineering degree, producing two conference presentations.

Mar 2018 – Nov 2019
Rosario, Argentina

PROFESSIONAL EXPERIENCE

- Data Engineer**
Abcam (a Danaher company)

Sep 2021 - Present
Amsterdam, The Netherlands
- **Partnered with data science teams to productise machine learning models**, translating research outputs into production-ready systems and embedding them into scalable data pipelines to support business and scientific use cases.
 - **Owned and delivered multiple internal data and platform initiatives end-to-end**, collaborating with cross-functional stakeholders (engineering, product, and business teams) to define requirements, align priorities, and drive projects from design through implementation and adoption.
 - **Led and contributed to strategic data initiatives**, including a period overseeing business intelligence operations and contributing to key projects migrating legacy systems to modern, maintainable, and scalable architectures.

SKILLS

Programming	Python, R, SQL, Java, L ^A T _E X
ML & NLP	PyTorch, Hugging Face Transformers, Tokenizers, TensorFlow, Scikit-learn, NLTK, spaCy, Stanza, NumPy, Pandas, SciPy
LLM APIs	OpenAI, LLaMA, Mistral
Visualisation	Matplotlib, Seaborn, Weights & Biases
Tools	Git, Linux, AWS, Jupyter, Pytest, CUDA
Languages	English (C2 – Bilingual), Spanish (Native), French (B2), Dutch (A1)